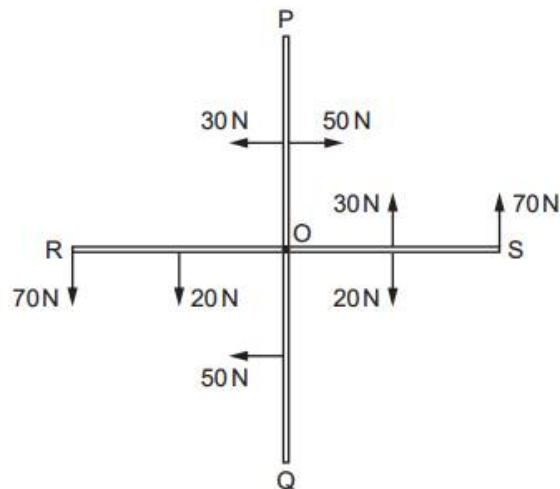


- 5 A rigid cross-shaped structure having four arms PO, SO, QO and RO, each 1.80 m long, is pivoted at O. Forces act on the ends of the arms and on the midpoints of the arms as shown.



What is the net moment about O?

- A** 54 N m **B** 108 N m **C** 216 N m **D** 432 N m

Ans: C

$$\text{scw} = (20+30+30) \times 0.9 + (70+70) \times 1.8 = 324 \text{ N m}$$

$$\text{scw} = (20+50+50) \times 0.9 = 108 \text{ N m}$$

$$\text{Net moments} = 216 \text{ N m}$$

- 6 Forces 5 N, 12 N and 12 N act at a point which is in equilibrium